

TENNIS MOBILE APP

Flutter-First Master Project Phases & AI Prompts

Mobile app first. Web administration and platform administration follow later.

Project Priority Override

The project will now begin with the PLAYER MOBILE APP only, built in Flutter. All planning, design-system work, information architecture, user journeys, domain modelling and implementation should prioritise the mobile player experience first. The Club/Community Admin web app and Platform Admin dashboard remain in the overall roadmap but are deferred until the Flutter mobile product has reached a stable milestone.

- Primary implementation target now: Flutter mobile app.
- Primary backend dependency: NestJS API and services needed by the mobile app.
- Next.js Club/Community Admin: deferred to a later phase.
- Next.js Platform Admin: deferred to a later phase.
- Tennis remains the primary product identity.
- Padel remains a secondary expansion and should not delay the Tennis mobile MVP.
- No global sport switcher.
- Primary colour remains #1C91D0.

Mobile-First Delivery Strategy

1. Phase M0 — Mobile Product Foundation
2. Phase M1 — Mobile Information Architecture & User Journeys
3. Phase M2 — Flutter Design System & App Shell
4. Phase M3 — Authentication & Adaptive Tennis Onboarding
5. Phase M4 — Personalised Home
6. Phase M5 — Player Discovery, Profiles & Connections
7. Phase M6 — Match Challenge, Scheduling & Messaging
8. Phase M7 — Match Scoring, Results, Ratings & Statistics
9. Phase M8 — Leagues, Seasons, Rounds & Standings
10. Phase M9 — Court & Club Discovery
11. Phase M10 — Learning, Skills, Practice & Progress
12. Phase M11 — Tennis News & Content
13. Phase M12 — Notifications, Profile, Settings & Safety
14. Phase M13 — Padel Expansion in Mobile
15. Phase M14 — Mobile Payments & Monetisation
16. Phase M15 — Analytics, Offline/Network Resilience & Observability
17. Phase M16 — QA, Accessibility, Performance & Security
18. Phase M17 — Beta, Store Readiness & Launch
19. Later Phase W1 — Club/Community Admin Web App
20. Later Phase W2 — Platform Admin Dashboard

Updated Master Starter Prompt — Flutter First

You are a Senior Product Designer, UX Architect, Product Strategist, Flutter Architect, Design Systems Lead, and Software Architect.

You are helping me plan and build a modern TENNIS-FIRST mobile platform.

IMPORTANT:

The PLAYER MOBILE APP is the FIRST and PRIMARY product to build.

Technology:

- Flutter: Player Mobile App
- NestJS: Backend/API required by the mobile app

Later, after the mobile product is stable:

- Next.js: Club / Community Admin Web App
- Next.js: Platform Admin Dashboard

Do NOT begin with the web applications.

Do NOT split attention equally across mobile and web.

Do NOT start building production screens before the mobile product foundation, journeys and design system are defined.

PRODUCT POSITIONING:

This is a Tennis-first platform.

Padel is a secondary expansion.

There is NO global sport switcher.

Padel may later be added through:

Profile → My Sports → + Add Padel

PRIMARY BRAND COLOUR:

#1C91D0

MOBILE PRODUCT PILLARS:

PLAY

COMPETE

DISCOVER

IMPROVE

FOLLOW

CONNECT

PRIMARY MOBILE LOOP:

Discover

→ Register

→ Adaptive Tennis Onboarding

→ Suggested Level

→ Personalised Home

- Find Player / Competition / Court
- Connect
- Challenge
- Schedule
- Play
- Record Result
- Confirm
- Update Rating & Stats
- Reflect
- Improve
- Play Again

FIRST DELIVERABLES:

1. Flutter mobile product architecture
2. Mobile information architecture
3. Mobile navigation
4. Complete mobile user journeys
5. Complete mobile screen inventory
6. Flutter design-system tokens
7. Flutter component inventory
8. Mobile domain/data requirements
9. NestJS API needs for mobile
10. Mobile MVP scope
11. Recommended Flutter implementation order

STOP after these are delivered.

Do not begin production feature implementation until approved.

Phase M0 — Mobile Product Foundation

Create the Flutter-first mobile product foundation.

Focus ONLY on the player mobile app and NestJS capabilities required to support it.

Deliver:

- Mobile product vision
- Mobile personas
- Jobs-to-be-done
- Mobile feature map
- Mobile P0/P1/P2 scope
- Mobile domain model
- Required NestJS services/endpoints at a capability level
- Mobile risks/dependencies
- Recommended Flutter build sequence

Do not design web admin products now.

Do not build screens yet.

STOP for review.

Phase M1 — Mobile Information Architecture & Journeys

Create the complete Flutter mobile information architecture and journey map.

Evaluate mobile top-level navigation:

Home

Play

Compete

Discover

Profile

Do not add a global sport switcher.

Create detailed flows for:

Registration

Adaptive Tennis onboarding

Player discovery

Connection request

Match challenge

Match scheduling

Court selection

Result submission

League registration

League participation

Court discovery

Learning

Practice

Progress

News

Notifications

Profile

Add Padel

Then produce the COMPLETE Flutter screen inventory with:

Purpose

Entry point

Primary action

Secondary actions

Data

States

Empty states

Error states

Connected screens

Do not create high-fidelity UI yet.

Phase M2 — Flutter Design System & App Shell

Build the Flutter design-system foundation and application shell.

Primary colour:

#1C91D0

Use:

Inter

8px spacing

WCAG 2.2 AA

Define Flutter:

ThemeData

ColorScheme

Theme Extensions

Typography

Spacing tokens

Radius tokens

Elevation

Motion

Icons

Responsive rules

Dark-mode strategy if included

Build reusable Flutter components for:

Buttons

Inputs

Search

Tabs

Navigation

Cards

Dialogs

Bottom sheets

Chips

Badges

Avatars

Player cards

Match cards

League cards

Court cards

Coach cards

News cards

Assessment questions

Rating cards

Statistics

Progress

Empty/loading/error states

Then create the Flutter app shell and navigation structure only.

Do not build full feature screens yet.

Phase M3 — Authentication & Adaptive Tennis Onboarding

Implement Flutter authentication and adaptive Tennis onboarding.

Flow:

Welcome

Register/Login

Verification

Basic Profile

Tennis Experience

Adaptive Tennis Assessment

Playing Preferences

Goals

Location

Club / Preferred Courts

Availability

Suggested Level

Confirm / Adjust

Profile Created

Personalised Next Actions

Assessment must branch dynamically.

Prefer behavioural questions over arbitrary 1–10 self-rating.

Assess:

Forehand

Backhand

Serve

Return

Net Play

Movement

Match Play

Store System Suggested Level and User Selected Level separately.

Optionally ask:

Do you also play Padel?

Yes / No / I'd like to learn

Do not interrupt the Tennis-first journey.

No global sport switcher.

Phase M4 — Personalised Home

Implement the Flutter Home experience.

Home must answer:

What should I do next?

Prioritise:

Urgent actions

Upcoming matches

League/round actions

Suggested opponents

Nearby courts

Development recommendations

Progress

News

Community updates

Create differentiated states for:

New user

Beginner

Active league player

Upcoming match

Unconfirmed result

Inactive player

Competitive player

Avoid a static dashboard.

Phase M5 — Player Discovery, Profiles & Connections

Implement Flutter player discovery, player profiles and connections.

Filters:

General location/distance

Rating/level

Availability

Singles/Doubles/Both

Social/Competitive/Both

Club

Preferred courts

Profile:

Photo

Name

General location

Club

Playing hand

Singles rating

Doubles rating

Stats

Recent form

Match history

Achievements

Connections

Availability summary

Development areas

Actions:

Connect

Accept

Decline

Pending

Remove

Block

Report

Protect privacy.

Phase M6 — Match Challenge, Scheduling & Messaging

Implement Flutter match challenge, structured scheduling and messaging.

Structured actions:

Challenge Player

Propose Time

Counter-propose

Suggest Court

Accept

Confirm

Reschedule

Cancel

Open Booking

Enter Result

Chat should complement the workflow, not replace it.

Support system messages for match-state changes.

States:

Proposed

Scheduling

Scheduled

Rescheduled

Cancelled

Expired

Handle edge cases and poor network states.

Phase M7 — Scoring, Results, Ratings & Statistics

Implement Flutter match scoring, result confirmation, disputes, ratings and statistics.

Flow:

Enter Score

Submit

Opponent Reviews

Confirm OR Dispute

Finalise

Update Rating

Update Stats

Update Standings

Optional Skill Reflection

Support:

Completed

Walkover

Retired

Cancelled

Disputed

Show:

Set scores

Winner

Date/time

Court

Opponent

Competition context

Recent form

Win/loss record

Match history

Keep Competitive Rating separate from Skill Development.

Phase M8 — Leagues, Seasons, Rounds & Standings

Implement the Flutter competition experience.

Support:

League discovery

League details

Rules

Season details

Registration

Waitlist

Enrolled state

Rounds

Opponents

Fixtures

Results

Standings

Announcements

Withdrawals

Walkovers

Disputes

Competition states:

Draft

Registration Open

Registration Closed

Scheduled

Active

Completed

Cancelled

Define how unfinished matches are handled when rounds/seasons close.

Phase M9 — Court & Club Discovery

Implement Flutter court and club discovery.

Include:

Map view

List view

Search

Filters

Court profile

Club profile

Directions

Call/Contact

External booking

Native booking placeholder if unavailable

Filters:

Distance

Surface

Indoor/Outdoor

Lighting

Public/Private

Amenities

Booking availability where known

Use Google Business information only where available and permitted.

Never fabricate availability.

Phase M10 — Learning, Skills, Practice & Progress

Implement Flutter Tennis learning and development.

Development dimensions:

Serve

Forehand

Backhand

Return

Net Play

Movement

Match Play

Create:

Skill Profile

Skill Detail

Goals

Lessons

Drills

Training Plans

Practice Log

Progress

Recommendations

Achievements/Streaks where appropriate

Connect:

Weakness/Goal

→ Recommended Drill/Lesson

→ Practice

→ Progress Update

Keep practice logging lightweight.

Phase M11 — Tennis News & Content

Implement Flutter Tennis news.

Categories:

Latest

Professional Tennis

Players

Tournaments

Local

Africa

Clubs

Community

Story card:

Headline

Publisher

Image where permitted

Short highlight

Publication date

Category

Source

Actions:

Read Highlight

Open Source

Save

Share

Follow Topic

Do not republish full third-party articles.

Prefer permitted APIs/RSS/licensed or public metadata sources.

Phase M12 — Notifications, Profile, Settings & Safety

Implement Flutter notifications, profile, settings and safety.

Notifications:

- Matches
- League rounds
- Results
- Connections
- Messages
- Competitions
- Learning
- News opt-in

Profile:

- Tennis profile
- Ratings
- Stats
- Achievements
- My Sports
- + Add Padel

Settings:

- Account
- Privacy
- Notification preferences
- Blocked users
- Help
- Data controls
- Logout

Safety:

- Block
- Report user
- Report message
- Report content
- Manage connection requests.

Phase M13 — Padel Expansion in Mobile

Extend the Flutter app with Padel as a secondary capability.

No global sport switcher.

Entry:

Profile

→ My Sports

→ + Add Padel

Add:

Padel profile

Adaptive Padel assessment

Padel rating

Padel match history

Padel stats

Padel preferences

Padel goals

Padel achievements

Expose Padel contextually in:

Player filters

Court filters

Competition filters

Learn

My Sports

Tennis remains the default.

Phase M14 — Mobile Payments & Monetisation

Define and implement mobile payment/entitlement flows required for the MVP and near-term roadmap.

Evaluate:

Free player tier

Premium player tier

League/event fees

Court booking payments

Coach payments if later required

Implement:

Entitlements

Upgrade/downgrade

Billing history

Receipts

Failed payments

Refund/cancellation states

Keep payment-provider architecture flexible.

Phase M15 — Analytics, Offline Resilience & Observability

Implement mobile analytics and resilience.

Track:

Registration

Assessment

Player discovery

Connections

Match challenge

Match scheduling

Match completion

League joins

Court views

Practice

Learning

Retention

Handle:

Poor network

Retry

Duplicate submissions

Optimistic updates where safe

Offline-friendly read states where practical

Crash/error reporting

API observability

Do not track unnecessary personal data.

Phase M16 — QA, Accessibility, Performance & Security

Create and execute the Flutter pre-release quality phase.

Accessibility:

WCAG 2.2 AA

Screen-reader labels

Dynamic text

Touch targets

Contrast

Reduced motion

Performance:

Cold start

List rendering

Images

Maps

Scrolling

API latency

Battery/network efficiency

Security:

Auth/session

Secure storage

PII

Input validation

Rate limiting

Abuse controls

QA:

Happy paths

Edge cases

Poor network

Duplicate taps

Interrupted onboarding

Result disputes

Competition close

Booking failures

Produce release-blocker criteria.

Phase M17 — Beta, Store Readiness & Launch

Prepare the Flutter mobile app for beta and launch.

Define:

Closed beta

Pilot players/clubs

Feedback capture

Bug triage

Support

Analytics dashboards

Launch metrics

App Store readiness

Google Play readiness

Privacy disclosures

Production monitoring

30/60/90-day optimisation plan

Prioritise outcomes:

Players find opponents

Players schedule matches

Players complete matches

Players join competitions

Players play more often

Players return to the app.

Phase W1 — Later — Club/Community Admin Web App

Only after the Flutter mobile app is stable, begin the Next.js Club/Community Admin product.

Use the approved mobile domain model and competition logic rather than inventing a separate system.

Do not start this phase before mobile milestones are approved.

Phase W2 — Later — Platform Admin Dashboard

Only after the Flutter mobile app and core club operations are stable, begin the Next.js Platform Admin dashboard.

Reuse shared entities, roles, audit rules and platform policies.

Do not start this phase during the initial Flutter build.

Implementation Principle

Build the Flutter player experience to a stable, usable milestone first. Do not let future admin portals dilute the mobile product's planning, UX quality, or implementation focus.